1 Quality-Diversity Placement Experiments

1.1 Quality-Diversity Archive Analysis

The original placement-generation experiments used CMA-MAE with a six-dimensional discretized descriptor space comprising position (x, y, z) and orientation $(\text{roll}, \text{pitch}, \text{yaw})$. Each object was evaluated at checkpoints of 540, 1080, 1620, and 2160 evaluations. The original successful-cell counts and QD scores were extracted from the archived experiment data.

Let N_{cells} denote the product of the six descriptor-grid dimensions, and let N_{success} denote the number of successful archive cells according to the original placement evaluation. The successful archive coverage is

$$C_{\text{success}} = \frac{N_{\text{success}}}{N_{\text{cells}}}. \tag{1}$$

The recorded successful-archive QD score uses a constant fitness shift of 10:

$$QD_{\text{success}} = \sum_{i \in \mathcal{A}_{\text{success}}} (f_i + 10). \tag{2}$$

Here, f_i is the archived fitness of successful cell i. This score depends on both the number of successful cells and their shifted fitness contributions.

All six objects showed nondecreasing successful-cell counts across the recorded checkpoints. Because the descriptor grids contain approximately 10^8 possible cells, the resulting fractional coverage is small. The grid resolutions also differ across objects, so absolute coverage percentages are not directly equivalent measures of placement-generation difficulty.

The plate is included in this archive analysis because original placement-generation data are available. It is excluded from the independent robot-grasp evaluation because no held-out plate grasp trials were performed. Archive success must not be interpreted as demonstrated robot-grasp success.

These results characterize archive population and recorded fitness, not comparative superiority over other QD algorithms. Such an inference would require matched algorithmic baselines and evaluation budgets.

2 Independent Simulated Grasp Evaluation

2.1 Experimental protocol

Five objects were evaluated using object-specific frozen grasp-planning protocols: spoon, banana, power drill, mug, and bowl. Development placements were excluded from the independent evaluations. For each held-out placement, the planner selected a grasp according to its predefined search and feasibility checks. Grasp execution was simulated using a Franka Panda manipulator in PyBullet.

The evaluation considered approach feasibility, bilateral finger contact, lifting, retention, table separation, and relative object–gripper slip, subject to the criteria defined for each object. The success definitions were not fully identical across objects: notably, the bowl protocol measured cumulative slip from the end of finger closure through lifting and retention, whereas the banana, power drill, and mug protocols measured slip during retention relative to the end of lifting. The spoon protocol also used different contact and success checks. Consequently, object-level rates are presented separately.

2.2 Object-level results

2.3 Comparison of placement-ranking methods

2.4 Selection overlap and interpretation

For spoon, Centrality-only and Height+Centrality selected identical placement sets, so their matching results do not constitute independent evidence. For banana, Centrality-only and Height+Centrality selected identical placement sets, so their matching results do not constitute independent evidence.

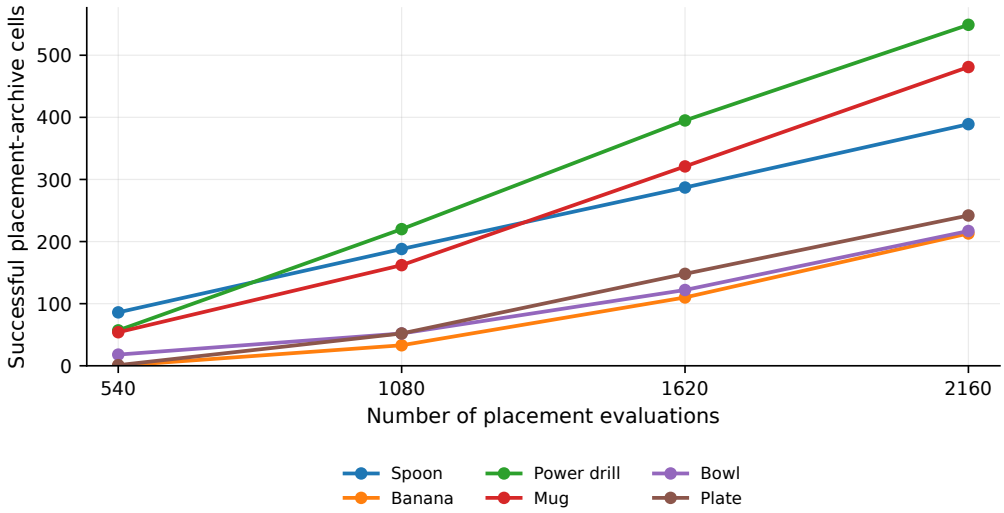


Figure 1: Growth in successful placement-archive cells for six objects under CMA-MAE at a fixed budget of 2160 evaluations. Archive success refers to the original placement criterion, not downstream robotic grasp success.

Table 1: Final original CMA-MAE placement-archive metrics after 2160 evaluations per object. Successful archive cells refer to the original placement criterion, not downstream robot-grasp success.

Object	Position bins per axis	Total grid cells	Successful cells	Coverage (%)	QD score
Spoon	60	216,000,000	389	0.0001801	3298.27
Banana	60	216,000,000	213	0.0000986	1694.51
Power drill	59	205,379,000	549	0.0002673	4513.79
Mug	52	140,608,000	481	0.0003421	4065.44
Bowl	56	175,616,000	217	0.0001236	1875.76
Plate	66	287,496,000	242	0.0000842	2027.50

For bowl, Centrality-only and Height+Centrality selected identical placement sets, so their matching results do not constitute independent evidence.

Across the evaluated objects, the GFI ranking did not consistently outperform the four baselines on downstream simulated grasp success. Several methods selected overlapping placements, and each placement had the same recorded outcome regardless of which ranking method selected it. The reported percentages are therefore descriptive comparisons of selection sets rather than estimates from independent randomized trials. They do not establish statistical superiority of a ranking method.

2.5 Failure analysis and limitations

The bowl exhibited especially limited performance under the fixed radial rim-grasp planner. Two of its 32 held-out placements succeeded. Thirteen had no statically feasible grasp within the predefined search and 17 failed during dynamic execution. Both successful placements were selected by the Uprightness-only ranking. These findings indicate a limitation of the tested planner and selection procedure, not a general impossibility of bowl grasping.

The plate was investigated only during grasp-planner development. No suitable opposing-finger

Table 2: Growth in successful placement-archive cells over the fixed CMA-MAE evaluation budget.

Object	540	1080	1620	2160
Spoon	86	188	287	389
Banana	0	33	110	213
Power drill	57	220	395	549
Mug	54	162	321	481
Bowl	18	52	122	217
Plate	1	52	148	242

Table 3: Independent simulated grasp outcomes by object. These rates use object-specific protocols and are not intended as a pooled benchmark.

Object	Successful	Evaluated	Success rate
Spoon	22	24	91.67%
Banana	21	29	72.41%
Power drill	34	35	97.14%
Mug	22	34	64.71%
Bowl	2	32	6.25%

configuration was demonstrated within the tested searches. Its 30 held-out placements were not evaluated; consequently, no plate success rate is reported and the plate is excluded from the independent grasp comparison.

The reported outcomes arise from simulated Panda executions rather than physical-robot experiments. Restricted grasp families, object-specific candidate generators, small method-selected sample sizes, selection overlap, and differences in success criteria limit generalization.

2.6 Implications for grasp-friendly placement

The experiments distinguish placement evaluation from downstream grasp execution. While quality-diversity methods can organize placements according to stability, diversity, and grasp-related attributes, the present independent execution results do not establish that GFI ranking consistently improves grasp success over the tested baselines. The relationship between placement descriptors and successful manipulation remains dependent on object geometry and grasp-planner capability.

Table 4: Simulated grasp success within the placements selected by each ranking method. Numerators denote successful placements and denominators denote selected held-out placements. Selection sets may overlap.

Object	GFI	Centrality	Height	Uprightness	Height+Centrality
Spoon	6/7 (85.7%)	6/7 (85.7%)	7/8 (87.5%)	9/9 (100.0%)	6/7 (85.7%)
Banana	6/9 (66.7%)	7/9 (77.8%)	6/10 (60.0%)	8/10 (80.0%)	7/9 (77.8%)
Power drill	10/10 (100.0%)	10/10 (100.0%)	8/9 (88.9%)	10/10 (100.0%)	10/10 (100.0%)
Mug	4/10 (40.0%)	7/9 (77.8%)	6/10 (60.0%)	7/10 (70.0%)	7/9 (77.8%)
Bowl	0/10 (0.0%)	0/10 (0.0%)	0/10 (0.0%)	2/9 (22.2%)	0/10 (0.0%)